

TDWI Data Management Maturity Model — Assessment Instrument

MIS774 Enterprise Data Management | Deakin University | Total marks: 100 (20 per dimension)

Student Information

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Unit	Organisation / Case Study	Assessor Role
	Organisation being assessed	Consultant

Instructions

Select the response that best describes the current state of the organisation you are assessing. Each dimension carries **20 marks**

. Your total score out of 100 maps to a TDWI maturity stage. Scores update automatically as you answer. Answer all 88 questions before printing or submitting.
Note: Information used in this assessment should be gathered through interviews, observation, and review of relevant organisational documentation.

Dimension 1: Organisation

Score: **6.0** / 20 | **Developing**

LEADERSHIP

Q1	Your organisation has identified an executive stakeholder(s) responsible for data strategy and use	2 marks
	No, but we are moving that way	✓
Q2	Executive leadership values data for decision-making	1.5 marks
	Disagree	✓
Q3	Executive leadership supports and communicates our data strategy	1.5 marks
	Disagree	✓

Q4 Executive leadership in my company has defined and enabled a data-driven culture

2 marks

No, but we are moving that way and leadership discusses being data-driven



CULTURE

Q5 Data literacy is a priority for my organisation

1.5 marks

Completely disagree



Q6 In my organisation, decisions are data-driven

2 marks

No, but we are using data in some strategic decisions



Q7 My organisation monetises its data

0.5 marks

Disagree



Q8 My organisation recognises the value of data

1 mark

Neutral — neither agree nor disagree



STRATEGY

Q9 My organisation has a data strategy

2 marks

No, but we are moving that way



Q10 The data strategy in my company is aligned and supports our organisational strategy (if no data strategy, select 'Completely disagree')

1.5 marks

Disagree



Q11 The data strategy is communicated and known across our organisation

1.5 marks

The data strategy is communicated, but not really understood



IMPACT

Q12 My organisation measures the business value of data

1 mark

Disagree



Q13 My organisation has established protocols for prioritising and managing data assets

1 mark

We do have some data assets prioritised and managed



DATA ETHICS

Q14 My organisation has established and communicated a code of ethics for data management

0.5 marks

Disagree



Q15 Data ethics accountability is established in my organisation

0.5 marks

Disagree



Dimension 2: Resources

Score: **5.1** / 20 | **Developing**

FUNDING

Q16 My organisation has invested in data management and governance

1.5 marks

Neutral — neither agree nor disagree



Q17 My organisation has allocated budget to enable the needed level of data management

1.5 marks

We do have some budget, but have gaps in data management funding



TALENT AND SKILLS

- Q18 My organisation regularly recruits resources with the required data management skills needed for growth 1.5 marks
- Disagree
- Q19 My organisation has the right skills to deliver data management as a core competency 1 mark
- No, we have data management skills, but they are traditional only (e.g. DBA, data modeler)
- Q20 My organisation has a strategy to enable the needed level of data management skill levels in the organisation 1.5 marks
- Completely disagree

ROLES AND RESPONSIBILITIES

- Q21 My organisation has defined IT data management roles and responsibilities 1.5 marks
- Yes, we have IT roles, but responsibilities are not clear tactically or strategically
- Q22 My organisation has defined business data management roles and responsibilities (e.g. data steward) 1.5 marks
- No, we don't have clear roles or responsibilities
- Q23 My organisation has defined executive data management roles and responsibilities 1.5 marks
- No, we don't have clear roles or responsibilities
- Q24 My company has a defined data governance organisation 1.5 marks
- Disagree
- Q25 It is clear how our organisational data management roles work together to provide business value 1 mark
- Disagree

EDUCATION

Q26 There are specific data management training courses for those in the lines of business, in addition to data management/governance professionals 1.5 marks

Disagree



Q27 The data management education available is adequate for our data management needs 1.5 marks

Completely disagree



LITERACY

Q28 My organisation has defined its data sources and constructs (including key systems, subject areas, and the rules that govern them) 1 mark

Yes, we have defined some data sources and some rules that govern the sources



Q29 My organisation has the knowledge and aptitude required to work with, interpret, and act on data 1 mark

Completely disagree



Q30 My organisation has knowledge of analytical methods and techniques to deliver business value (predictive, prescriptive, or machine learning models) 1 mark

Yes, we have knowledge and use some analytical methods



Dimension 3: Architecture

Score: **12.6** / 20 | Managed

DIVERSITY, VOLUME, SPEED

Q31 Our architecture enables real-time data ingestion (i.e., streaming or continuous) 1 mark

Yes, the architecture supports real-time data ingestion with no issues



Q32 My company's architecture scales to support data volume requirements 1 mark

Neutral — neither agree nor disagree



Q33 My company's architecture supports the acquisition, storage, and use of structured AND unstructured data 1 mark

Yes, we can support structured and some unstructured data types

Q34 In our architecture, there are no discernible barriers to volume, speed, or diversity — data requirements are consistently met 1 mark

We meet most or all of the volume, speed, or diversity requirements

DATA ACCESS

Q35 Our architecture supports connection to data from different locations and devices 1 mark

We can support different devices and locations, but have some architectural limitations

Q36 How accessible is your data that is required for business value? 1.5 marks

Data is accessible to a limited number of users

Q37 Data is accessible at the right time (with the right level of latency) in my company 1 mark

Neutral — neither agree nor disagree

Q38 My company's architecture supports the ability to query the data sources and volumes needed for business value 1 mark

We meet most or all of the volume, speed, or diversity requirements

DATA INTEGRATION AND MANAGEMENT

Q39 Our data architecture enables the ease of data integration and use of relevant data 2 marks

Data integration is getting better, but still takes longer than we need

Q40 Our data architecture includes data pipelines that are scalable and automated 2 marks

Data pipelines are present; automation is possible and existing pipelines perform fairly well

- Q41 We use multiple data delivery methods besides ETL (e.g., CDC, data virtualisation, ESB, event bus, etc.) 0.5 marks
- We use basic ETL and change data capture (CDC) ✓
- Q42 Our architecture includes data management tools such as data catalogs, lineage, metadata, and code repositories that are accessible and integrated 0.5 marks
- We have data management tools, but not all listed and accessibility is a challenge ✓

TECHNICAL INFRASTRUCTURE

- Q43 Our technical infrastructure enables predictive and prescriptive model building against large volumes of data 1 mark
- Our architecture can support predictive or prescriptive models, but performance is a problem against large volumes ✓
- Q44 Our technical infrastructure supports analytics model operations 1 mark
- Completely agree ✓
- Q45 Our technical infrastructure supports data pipeline and other automation (e.g., automated deployment, scheduling, testing) 1 mark
- Neutral — neither agree nor disagree ✓
- Q46 Our technical infrastructure supports microservices (i.e., an architectural style that develops a single application as a set of small services, communicating via lightweight protocols or APIs) 0.5 marks
- Agree ✓
- Q47 How much time does your organisation spend on operations tasks such as back-up, patch updates, data recovery, data movement, etc.? 1 mark
- We spend time doing this although we are trying to automate some of it ✓

ARCHITECTURE ALIGNMENT

Q48 Our data and technical architecture supports the delivery of the organisation's strategic goals

2 marks

Completely agree



Dimension 4: Data Life Cycle

Score: **4.0** / 20 | **Developing**

SCOPE OF CAPABILITIES

Q49 Data acquisition/integration is simple and scalable in my organisation

1.5 marks

Neutral — neither agree nor disagree



Q50 Data is sharable and accessible across my organisation

1.5 marks

Disagree



Q51 Data assets are reusable across data and analytics projects in my organisation

1 mark

We don't regularly reuse data assets across projects



Q52 Data security and privacy requirements are consistently met in my organisation

1.5 marks

Completely disagree



AUTOMATION AND AUGMENTATION

Q53 Software components within the data ecosystem consistently scale and perform according to expectations in my organisation

1 mark

Agree



Q54 In my company, project deployment is automated

1 mark

We have some automation, but still use manual deployment



Q55 In my company, self-service capabilities are available for the data consumers

1 mark

We have self-service for some users or some parts of the organisation



DEPLOYMENT AND DELIVERY APPROACHES

Q56 My organisation has methodologies that foster collaboration between everyone handling data (developers, engineers, data scientists, analysts, and business users)

1 mark

We have methodologies, but none that foster collaboration



Q57 My organisation uses agile methodology for the delivery of data assets

1 mark

We have methodologies, but they are not focused on agile practices



Q58 My organisation has a DataOps or DevOps team that fosters reusability, automation, and continuous improvement

1.5 marks

We don't have a DevOps or DataOps team



Q59 In my organisation, data projects are consistently delivered and deployed successfully (stakeholder expectations are consistently met)

1 mark

Agree



CURATION

Q60 In my organisation, metadata management is integrated into the data lifecycle

1 mark

We have metadata, but it is not integrated into the lifecycle



Q61 Data catalogs are available and well maintained in my organisation

1 mark

We don't have data catalogs



Q62 In my organisation, data curation (maintaining a data catalog, lineage, metadata, etc.) is collaborative across technical and business teams

0.5 marks

Data curation is not yet a focus for our organisation



Q63 In my organisation, data curation is a shared and collaborative function across business and technical teams (overall assessment)

1 mark

Disagree



VALUE OF ANALYTICS

Q64 My organisation can measure the value of analytics

1 mark

Completely disagree



Q65 KPIs for measuring analytics success are known and communicated in my organisation

1 mark

We don't have KPIs to measure analytics value



STORAGE AND OPERATIONS

Q66 In my organisation, storage options scale and are accessible

0.5 marks

Disagree



Q67 My organisation has incorporated cloud storage into the ecosystem

1 mark

We don't use cloud storage



Dimension 5: Governance

Score: **2.7** / 20 | Nascent

SECURITY AND PRIVACY

Q68 My organisation makes use of encryption where needed for data in the ecosystem

0.75 marks

Neutral — neither agree nor disagree



- Q69 My organisation has the right processes in place to keep our data secure, no matter where it resides 1 mark
- Completely disagree ▼
- Q70 My organisation has identified, supported, and applied data obfuscation techniques 0.5 marks
- Disagree ▼
- Q71 My organisation archives and deletes data per regulatory/compliance requirements 0.75 marks
- Disagree ▼

DATA GOVERNANCE PROCESSES AND TOOLING

- Q72 My organisation has good tools and techniques in place to assess and consistently monitor the quality and accuracy of data in our ecosystem 1 mark
- Completely disagree ▼
- Q73 My organisation has metadata available to users and applications that is used to promote discovery and reuse 0.75 marks
- Disagree ▼
- Q74 My organisation has defined information sensitivity classification which is supported in the data catalogs and used to control data access and use 0.75 marks
- Disagree ▼
- Q75 In my company, data assets are mapped to an appropriate retention schedule that complies with regulatory/compliance requirements 0.5 marks
- Data assets do not currently have a retention schedule ▼

GOVERNANCE ROLES

- Q76 My organisation has an executive role for data governance identified and in place (e.g. Chief Data Officer) with accountability for data governance 2 marks
- No one is assigned to this role at this time ▼

Q77 My organisation has data owners for priority data subject areas identified and engaged in data governance (data owners have leadership accountability for data assets) 2 marks

No one is assigned to the role at this time



Q78 My organisation has data stewards for priority data subject areas identified and engaged in data governance (data steward focuses on data quality and fitness for purpose) 2 marks

No one is assigned to this role at this time



Q79 My organisation has technical data custodians who apply and enforce data integrity, security, and quality requirements to the data ecosystem 1 mark

Completely disagree



MODEL GOVERNANCE PROCESSES AND TOOLS

Q80 My organisation has data modelling tools that support data design and development activities 0.75 marks

Disagree



Q81 My organisation has preferred data standards for how data is modelled and exchanged 1 mark

We are starting to do that now



Q82 My organisation has modelling processes that support multiple design approaches (e.g. top-down and bottom-up, agile, etc.) 0.75 marks

Neutral — neither agree nor disagree



QUALITY

Q83 My organisation has data quality rules and measurements being managed according to an agreed standard 1 mark

Disagree



- Q84 My organisation has data quality issue management policies, standards and procedures that apply across on-premises and cloud environments 1 mark
- Completely disagree
- Q85 My organisation leverages data quality networks to enhance/ensure data quality 0.75 marks
- Disagree
- Q86 My organisation has management-level dashboards that report on data quality KPIs 0.75 marks
- Completely disagree

SOVEREIGNTY

- Q87 My organisation has requirements for managing data sovereignty and cross-border data movement risks defined 0.5 marks
- Completely disagree
- Q88 My organisation has controls for data sovereignty and cross-border data movement risk in place 0.5 marks
- Disagree

Assessment Summary Report

30.4
TOTAL SCORE / 100
Developing

DIMENSION	SCORE / 20	PROGRESS	MATURITY STAGE
 Dimension 1: Organisation	6.0 / 20		Developing
 Dimension 2: Resources	5.1 / 20		Developing
 Dimension 3: Architecture	12.6 / 20		Managed
 Dimension 4: Data Life Cycle	4.0 / 20		Developing

Dimension		Score / 20	Progress	Maturity Stage
 Dimension 5: Governance		2.7 / 20		Nascent
Overall Total		30.4 / 100		Stage 2 — Developing
Maturity Stage Reference (per dimension, out of 20)				
Stage 1 — Nascent (0–4) Stage 2 — Developing (4.1–8) Stage 3 — Established (8.1–12) Stage 4 — Managed (12.1–16) Stage 5 — Optimised (16.1–20)				